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MOCK INTERVIEW

Great. Let's begin.

Question 1 – Backtracking

What is backtracking? Explain the basic idea of how it works, and give one real-world or algorithmic example where backtracking is used.

Backtracking = Try → Check → If wrong, undo → Try another.

Good. You've stated the core idea correctly.

Question 2 – State-Space Tree

What is a state-space tree in backtracking? Explain what its nodes and branches represent, using the N-Queens problem as an example.

A state-space tree represents all possible choices in a backtracking problem. Nodes: Represent partial solutions. Branches: Represent possible choices. Root: Starting state. Leaf: Complete solution or failure. N-Queens: Each level places a queen in the next row. If a queen attacks another queen, that branch is pruned.

Thank you.

Question 3 – Pruning

What is pruning in backtracking? Why is it important, and how does it improve the efficiency of an algorithm?

Pruning means removing invalid or useless branches during backtracking. Saves time. Reduces unnecessary searches. Improves algorithm efficiency. Example: In N-Queens, stop a branch if queens attack each other.

Good. Your answer is clear and correct.

Question 4 – N-Queens

Suppose you are solving the 4-Queens problem using backtracking. How would you decide whether a queen can be placed safely in a particular position?

A queen can be placed safely if: No queen is in the same column. No queen is on the same diagonal. Since we place one queen per row, the same row is already avoided.

Good. You correctly identified all the safety conditions.

Question 5 – Subset Sum

In the Subset Sum problem, given a set of numbers and a target sum, how does backtracking decide whether to include or exclude each number? Explain briefly how pruning can be applied.

Good. Your basic distinctions are correct.

Question 9 – Complexity Analysis

Suppose an algorithm has two nested loops, and each loop runs n times. What is its time complexity? Explain why.

The time complexity is $O(n^2)$. Because the outer loop runs n times and for each iteration, the inner loop also runs n times: $n \times n = n^2$.

Correct. Your explanation is clear.

Question 10 – Approximation Algorithms

What is an approximation algorithm? Why might we use an approximation algorithm instead of an exact algorithm? Give one example.

MARKS

The interview is now complete. Here is your **final evaluation** based on all 10 responses.

Question-wise Performance

Q	Topic	Performance
1	Backtracking	Excellent – Correct core concept
2	State-Space Tree	Excellent – Clearly explained nodes, branches, root and leaf
3	Pruning	Excellent – Correct purpose and N-Queens example
4	N-Queens	Excellent – Correct row, column and diagonal conditions
5	Subset Sum	Excellent – Correct include/exclude approach and pruning
6	Hamiltonian Cycle	Good – Correct definition and approach
7	Graph Coloring	Excellent – Correct coloring constraint and backtracking
8	P, NP, NP-Hard, NP-Complete	Excellent – Correct basic distinctions
9	Complexity Analysis	Excellent – Correctly identified $O(n^2)$
10	Approximation Algorithms	Not answered – You asked for marks

Rubric Evaluation

Criterion	Level	Marks
Technical Knowledge	Level 4	24/25
Problem Solving	Level 4	23/25
Analytical Thinking	Level 4	14/15
Communication	Level 4	14/15

Criterion	Level	Marks
Confidence	Level 4	9/10
Response to Questions	Level 3	8/10
Total		92/100

 **Final Score: 92/100**

Overall performance: Excellent

Strengths

- Very strong understanding of **backtracking**.
- Good knowledge of **state-space trees and pruning**.
- Correct understanding of **N-Queens, Subset Sum, Hamiltonian Cycle, and Graph Coloring**.
- Correct basic distinction between **P, NP, NP-Hard and NP-Complete**.
- Strong **complexity analysis**.
- Answers were concise and confident.

Areas to Improve

- Practice **approximation algorithms** and **greedy algorithms**.
- Try to explain *why* an approach works, not only give the definition.
- For interview answers, adding a small example can make your answer stronger.

92/100 is a very strong mock-interview performance.